System Architecture & Database Design – ScrollControl

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1. System Architecture

1.1 Overview

ScrollControl follows a **mobile-first, client-heavy architecture** with optional backend support.

- Most logic runs **on-device (Flutter app)**
- Backend (Laravel) is **optional for sync and future features**
- Designed to be **privacy-first and scalable**

1.2 High-Level Architecture

User

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Flutter App (Mobile/Web)

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├— UI Layer

├— Business Logic Layer

├— Local Storage (SQLite)

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Optional Backend (Laravel API)

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└— Database (MySQL)

1.3 Architecture Components

1.3.1 Frontend (Flutter App)

Handles:

- UI rendering
- Focus timer
- App usage tracking (Android)
- Smart interrupt logic
- Local data storage

1.3.2 Local Storage (SQLite)

Purpose:

- Store user data locally
- Ensure offline functionality
- Improve performance

Stores:

- Focus sessions
- App usage logs
- Settings

1.3.3 Backend (Laravel API – Optional)

Used for:

- User authentication (future)
- Data synchronization across devices
- Analytics (future)

1.3.4 External Interfaces

- Android Usage Stats API
- Notification system
- System permissions manager

1.4 Data Flow

Scenario: Usage Tracking + Interrupt

1. User opens another app (e.g. Instagram)
2. Flutter app monitors usage via Android API
3. Usage exceeds threshold (e.g. 5 minutes)
4. App triggers notification
5. User responds:
    - Continue
    - Start focus session
6. Data stored locally

Scenario: Focus Session

1. User starts session
2. Timer runs in background
3. On completion:
    - Session saved locally
    - Stats updated

1.5 Design Principles

- **Local-first**: Data stays on device
- **Modular**: Easy to extend features
- **Permission-safe**: Minimal system access
- **Scalable**: Backend can be added later

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2. Database Design

2.1 Overview

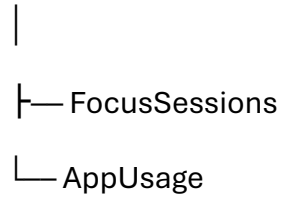
Database is split into:

- **Local SQLite database (primary)**

- **Remote MySQL database (optional)**

2.2 Entity Relationship Overview

User (optional)



2.3 Tables (Local Database)

2.3.1 FocusSessions Table

Field Name	Type	Description
id	INTEGER	Primary key
duration	INTEGER	Session length (in minutes)
start_time	DATETIME	When session started
end_time	DATETIME	When session ended
created_at	DATETIME	Record creation timestamp

2.3.2 AppUsage Table

Field Name	Type	Description
id	INTEGER	Primary key
app_name	TEXT	Name of application
package_name	TEXT	Unique app identifier
usage_time	INTEGER	Time spent (in seconds/minutes)
date	DATE	Usage date

Field Name	Type	Description
created_at	DATETIME	Record creation timestamp

2.3.3 Settings Table

Field Name	Type	Description
id	INTEGER	Primary key
tracking_enabled	BOOLEAN	Enable/disable tracking
interrupt_time	INTEGER	Threshold (minutes)
notifications_on	BOOLEAN	Notification toggle
created_at	DATETIME	Record creation timestamp

2.4 Optional Tables (Backend Database)

2.4.1 Users Table

Field Name	Type	Description
id	INTEGER	Primary key
email	TEXT	User email
password	TEXT	Hashed password
created_at	DATETIME	Timestamp

2.4.2 Synced Focus Sessions

Field Name	Type	Description
id	INTEGER	Primary key
user_id	INTEGER	Foreign key
duration	INTEGER	Session duration

Field Name	Type	Description
date	DATE	Session date

2.4.3 Synced App Usage

Field Name	Type	Description
id	INTEGER	Primary key
user_id	INTEGER	Foreign key
app_name	TEXT	App name
usage_time	INTEGER	Time spent
date	DATE	Usage date

2.5 Relationships

- One user → many focus sessions
- One user → many app usage records

2.6 Data Storage Strategy

Local First

- All data stored on device
- Ensures:
    - Privacy
    - Offline access
    - Faster performance

Optional Sync

- User opts in
- Data synced to backend
- Used for:

    - Multi-device support
    - Analytics

2.7 Indexing & Optimization

- Index on:
    - date (for quick stats retrieval)
    - package_name (for app tracking)

2.8 Data Retention Strategy

- Store recent data (e.g. last 30–90 days)
- Older data can be:
    - Aggregated
    - Deleted to save space

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3. Summary

The architecture ensures:

- Efficient real-time tracking
- Minimal reliance on backend
- Compliance with privacy standards
- Scalability for future growth
